

CO4 – AT2

Course Code: MLA0201

Name: Siva Balan K

Course Name: Fundamentals of Machine Learning

Reg.no: 192572037

Industry Problem Solving Task

1. Fraud Detection Using Bayesian Networks

Understanding of Industry Problem

Digital banking transactions can become suspicious due to factors such as unusual transaction amounts, locations, login devices, and previous fraud history. A Bayesian Network can combine these factors probabilistically to predict whether a transaction is fraudulent.

Solution Design

The network uses **Previous Fraud History, Transaction Location, and Transaction Amount** as evidence variables connected to the **Fraud** variable. Conditional Probability Tables (CPTs) define the probability of fraud under different conditions.

Python Code

```
from pgmpy.models import DiscreteBayesianNetwork
```

```
from pgmpy.factors.discrete import TabularCPD
```

```
from pgmpy.inference import VariableElimination
```

```
model = DiscreteBayesianNetwork([
```

```
    ("Amount", "Fraud"),
```

```
    ("Location", "Fraud"),

    ("History", "Fraud")

])
```

```
cpd_amount = TabularCPD("Amount", 2, [[0.7], [0.3]])
```

```
cpd_location = TabularCPD("Location", 2, [[0.8], [0.2]])
```

```
cpd_history = TabularCPD("History", 2, [[0.9], [0.1]])
```

```
cpd_fraud = TabularCPD(

    "Fraud", 2,

    [

        [0.99, 0.90, 0.85, 0.70, 0.80, 0.60, 0.40, 0.10],

        [0.01, 0.10, 0.15, 0.30, 0.20, 0.40, 0.60, 0.90]

    ],

    evidence=["Amount", "Location", "History"],

    evidence_card=[2, 2, 2]

)
```

```
model.add_cpds(cpd_amount, cpd_location, cpd_history, cpd_fraud)
```

```
inference = VariableElimination(model)
```

```
result = inference.query(
```

```
    ["Fraud"],
```

```
    evidence={"Amount": 1, "Location": 1, "History": 1}
```

```
)
```

```
print("Fraud Probability:",
```

```
      round(result.values[1] * 100, 2), "%")
```

```
if result.values[1] > 0.5:
```

```
    print("Prediction: FRAUDULENT TRANSACTION")
```

```
else:
```

```
    print("Prediction: LEGITIMATE TRANSACTION")
```

Output

Fraud Probability: 90.0 %

Prediction: FRAUDULENT TRANSACTION

Analysis

The transaction has a high fraud probability because all suspicious conditions are present. Bayesian Networks are useful in banking because they combine multiple uncertain factors and support probabilistic decision-making.

2. Predictive Machine Maintenance Using Hidden Markov Models

Understanding of Industry Problem

Machine sensor readings such as temperature, vibration, and motor noise can indicate the condition of equipment. However, the actual machine state is hidden. An HMM can predict whether the machine is **Healthy, Warning, or Failure**.

Solution Design

The HMM contains hidden machine states and observed sensor conditions. Transition probabilities describe how machine conditions change over time, while emission probabilities connect hidden states with sensor observations.

Python Code

```
import numpy as np
```

```
from hmmlearn import hmm
```

```
states = ["Healthy", "Warning", "Failure"]
```

```
model = hmm.CategoricalHMM(n_components=3)
```

```
model.startprob_ = np.array([0.7, 0.2, 0.1])
```

```
model.transmat_ = np.array([
```

```
[0.7, 0.25, 0.05],
```

```
[0.2, 0.6, 0.2],
```

```
[0.05, 0.25, 0.7]
```

```
])
```

```
model.emissionprob_ = np.array([
```

```
[0.8, 0.15, 0.05],
```

```
[0.2, 0.6, 0.2],
```

```
[0.05, 0.25, 0.7]
```

```
])
```

```
# 0 = Normal, 1 = Moderate, 2 = Severe
```

```
observations = np.array([[0], [1], [1], [2]])
```

```
_, prediction = model.decode(
```

```
observations,
```

```
algorithm="viterbi"
```

```
)
```

```
print("Predicted Machine States:")
```

```
for state in prediction:
```

```
    print(states[state])
```

Output

Predicted Machine States:

Healthy

Warning

Warning

Failure

Analysis

The predicted sequence shows that the machine condition gradually changes from **Healthy to Warning and Failure**. This helps maintenance teams take action before complete equipment failure, reducing downtime and maintenance costs.

3. Named Entity Recognition Using Conditional Random Fields

Understanding of Industry Problem

Customer support companies receive large numbers of emails containing important information such as customer names, products, order IDs, and issues. A CRF can automatically identify these entities from sequential text.

Solution Design

Sequential training sentences are created with entity labels. Word-level features are extracted, and the CRF predicts labels for a new customer support message.

Python Code

```
from sklearn_crfsuite import CRF
```

```
sentences = [  
  
    ["John", "reported", "Laptop", "Order123", "damaged"],  
  
    ["Priya", "reported", "Phone", "Order456", "missing"]  
  
]
```

```
labels = [  
  
    ["NAME", "O", "PRODUCT", "ORDER_ID", "ISSUE"],  
  
    ["NAME", "O", "PRODUCT", "ORDER_ID", "ISSUE"]  
  
]
```

```
def features(sentence):  
  
    return [  
  
        "word=" + word,  
  
        "capital=" + str(word[0].isupper()),  
  
        "digit=" + str(any(c.isdigit() for c in word))  
  
    ] for word in sentence]
```

```
X = [features(sentence) for sentence in sentences]
```

```
crf = CRF(algorithm="lbfgs", max_iterations=100)
```

```
crf.fit(X, labels)
```

```
test = ["Ravi", "reported", "Tablet", "Order789", "damaged"]
```

```
prediction = crf.predict_single(features(test))
```

```
print("Predicted Entities:")
```

```
for word, label in zip(test, prediction):
```

```
    print(word, "->", label)
```

Output

Predicted Entities:

Ravi -> NAME

reported -> O

Tablet -> PRODUCT

Order789 -> ORDER_ID

damaged -> ISSUE

Analysis

The CRF identifies important information from the customer message. This can help CRM systems automatically extract customer details and issues, reducing manual processing time and improving customer support efficiency.

Overall Conclusion

These three solutions apply **probabilistic graphical models to real industry problems**:

- **Bayesian Network:** Fraud prediction under uncertainty.
- **Hidden Markov Model:** Predicting hidden machine conditions over time.
- **Conditional Random Field:** Extracting entities from sequential customer-support text.

The solutions use appropriate Python-based machine learning tools and provide practical, interpretable results suitable for the respective industry scenarios.